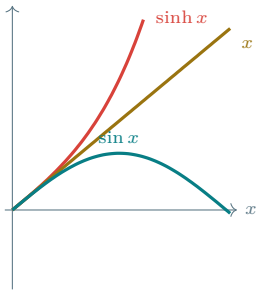
FUNCTIONS LADDER 7 OF 10

One equation, $y'' = \pm y$

The sign of the constant decides whether solutions grow, oscillate or run straight, and the characteristic equation finds them all.

1 / THE RULE AND THE PICTURE

Trying $y = e^{rx}$ in $y'' = ky$ gives the characteristic equation $r^2 = k$



solutions with $y(0) = 0$, $y'(0) = 1$: $\sinh x$, x and $\sin x$

Substituting $y = e^{rx}$ gives $r^2 e^{rx} = k e^{rx}$, and cancelling e^{rx},

$$\boxed{r^2 = k}$$

Two real roots give growth and decay, two imaginary roots give oscillation, and a double root gives a straight line. Every second-order linear equation with constant coefficients is solved by the same trick.

2 / THE THREE CASES

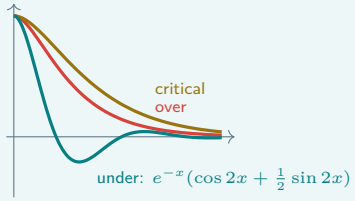
Six familiar functions are the solutions of three equations

equation	characteristic equation	roots	exponential solutions	real forms
$y'' = y$	$r^2 = 1$	$r = 1,\ -1$	$e^x,\ e^{-x}$	$\cosh x,\ \sinh x$
$y'' = -y$	$r^2 = -1$	$r = i,\ -i$	$e^{ix},\ e^{-ix}$	$\cos x,\ \sin x$
$y'' = 0$	$r^2 = 0$	$r = 0$ twice	$1,\ x$	$1,\ x$

The general solution is a combination of the two basic ones, for example $y = A\cos x + B\sin x$ for $y'' = -y$, and the two constants are fixed by $y(0)$ and $y'(0)$: choosing $y(0) = 1$, $y'(0) = 0$ picks $\cosh x$, $\cos x$ and 1, and $y(0) = 0$, $y'(0) = 1$ picks $\sinh x$, $\sin x$ and x.

3 / ADDING FRICTION / THE THREE NUMBER TYPES

For $y'' + by' + cy = 0$ the characteristic equation $r^2 + br + c = 0$ has three possible kinds of root



The discriminant $b^2 - 4c$ decides: positive gives two real roots and overdamped decay, such as $y'' + 3y' + 2y = 0$ with $r = -1, -2$; zero gives the critical case $(A + Bx)e^{-x}$ for $y'' + 2y' + y = 0$; negative gives complex roots $-\frac{b}{2} \pm i\omega$ and damped oscillation, as for $y'' + 2y' + 5y = 0$ with $r = -1 \pm 2i$ and $y = e^{-x}(A\cos 2x + B\sin 2x)$.

4 / PHYSICAL EXAMPLES

A mass on a spring oscillates, and an inverted pendulum falls exponentially

For $my'' = -ky$ with $m = 0.5$ kg and $k = 20$ N/m, $r^2 = -40$ so $\omega = \sqrt{40} = 6.32$ rad/s and the period is $2\pi/\omega = 0.993$ s.

A pendulum balanced upside down obeys $\theta'' = +\frac{g}{L}\theta$ for small angles, so $r = \pm\sqrt{g/L} = \pm 3.13$ per second for $L = 1$ m, and an error doubles every $\ln 2/3.13 = 0.22$ s, which is why balancing needs fast feedback.

5 / THE SAME IDEA FOR SEQUENCES

The Fibonacci recurrence has characteristic equation $r^2 = r + 1$

Trying $F_n = r^n$ in $F_{n+1} = F_n + F_{n-1}$ gives $r^2 = r + 1$, with roots $\varphi = 1.618$ and $-1/\varphi = -0.618$, so $F_n = A\varphi^n + B(-1/\varphi)^n$, which is the Binet formula.

Differential equations use the trial solution e^{rx} and recurrences the trial solution r^n, and in both cases a quadratic in r determines the behaviour.

ASIDE / THE DIFFERENCE

Real roots give monotone behaviour and complex roots give rotation; for equations the real part of the root sets the growth rate, and for recurrences its modulus does.

LINKS TO OTHER SHEETS

The recurrence version is in *The characteristic equation beneath five presentations*; the previous sheet is *Euler's formula and the complex exponential* and the next is *Taylor series*.

Convention. r is a root of the characteristic equation; ω an angular frequency; b, c constants; y' and y'' derivatives with respect to x.